



School of Computer Science

A Level 1 Module COMP1038 Autumn 2025/26

Programming and Algorithms

Instructions:

Time allowed: One Hour

Answer ALL questions

Candidates may complete the front cover of their answer book and sign their desk card. **DO NOT turn the examination paper over until instructed to do so.**

Permitted resources:

Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination.

No calculators are permitted in this examination.

Prohibited resources:

All other dictionaries, including subject specific translation dictionaries and electronic dictionaries. Electronic devices capable of storing and retrieving text.

Exam Papers must not be removed from the examination venue.

Question 1: Choose the correct answer to the following questions**[5 marks]**

- i. What will be the output of the following code snippet?

```
#include <stdio.h>
int main() {
    int a=5, b=2;
    printf("%d", a/b);
    return 0;
}
```

- a. 2.5
- b. 2
- c. 3
- d. Compilation error

[1 Mark]

- ii. Which of the following statements about the `static` keyword in C is correct?

- a. It limits the scope of a variable to the current file only.
- b. It limits the scope of a variable to the current function and maintains its value between function calls.
- c. It makes a function visible to other files.
- d. It allocates memory on the heap.

[1 Mark]

- iii. What is the output of the following code?

```
#include <stdio.h>
int main() {
    char s[] = "Hello";
    int i = 0;
    while(s[i]) {
        printf("%c", s[i++]);
    }
    return 0;
}
```

- a. H
- b. H e l l o
- c. Hello
- d. Compilation error

[1 Mark]

- iv. Given the following function, what will be printed when `func(5)` is called?

```
void func(int n) {
    if (n > 0) {
        func(n - 1);
        printf("%d", n);
    }
}
```

- a. 5 4 3 2 1
- b. 1 2 3 4 5
- c. 5
- d. 1 5

[1 Mark]

- v. Consider the following code for pointer arithmetic. What is the output?

```
#include <stdio.h>
int main() {
    int arr[] = {1, 2, 3, 4, 5};
    int *p = arr;
    printf("%d", *(p+3));
    return 0;
}
```

- a. 1
- b. 3
- c. 4
- d. 5

[1 Mark]

Question 2: What is a NULL pointer in the C programming language? Is it equivalent to an uninitialized pointer? Provide code examples for both.

[5 Marks]

Question 3: Error Identification and Correction

The following C program is designed to count and print the number of even numbers in an array entered by the user. However, it contains several errors. Identify and correct these errors. Explain each correction that you made.

[5 Marks]

```
1  #include <stdio.h>

2  int main() {
3  int n, i, count;
4  int arr[10];
5  printf("Enter number of elements (not more than 10): ");
6  scanf("%d", n);

7  printf("Enter %d integers:\n", n);
8  for (i = 1; i <= n; i++) {
9      scanf("%d", arr[i]);
10 }

11 for (i = 0; i < n; i++);
12     if (arr[] % 2 == 0)
13         count++;

14 printf("Number of even numbers: %d", count);
15 return 0;
16 }
```

Question 4: Array Manipulation and Counting

[7 Marks]

Problem Statement: Write a C program that reads an integer array of size n (where $2 \leq n \leq 20$) from the user, then determines and displays *all* unique pairs of elements whose sum is equal to a given target value (also input by the user).

For each pair, print the indices of the two elements (indices start at 0) and their values. If no such pairs are found, print an appropriate message.

Question 5: Explain with an example how a nested structure (structure within a structure) can be defined and used in C to represent hierarchical data.

[3 Marks]

--- End ---